SENTINELX // AUTONOMOUS AI SOC & EDR PLATFORM

Academic Research Foundations & Algorithmic Implementation Report

Project Domain: Cybersecurity, Graph Neural Networks (GNN) & Zero-Day Detection

Date: October 2026

Focus: Mapping Theoretical Academic Literature to Production Implementation

Scope: 4th Year B.Tech / BE Major Project

1. Master Research Paper & Standards Implementation Matrix

SentinelX rejects synthetic placeholder metrics. The table below maps each peer-reviewed paper and formal cybersecurity standard directly to its mathematical foundation and exact codebase location:

Research Paper / Standard	Authors & Venue	Theoretical Core Concept	Codebase Location in SentinelX
1. GraphSAGE **Inductive Representation Learning**	Hamilton, Ying, Leskovec *NeurIPS 2017*	Inductive neighborhood aggregation over k-NN flow graphs for Zero-Day novel attack discovery.	`ml/graphsage/model.py` `ml/graphsage/train.py`
2. CIC-IDS2017 / 2018 **Modern Attack Dataset & Flow Metrics**	Sharafaldin, Lashkari, Ghorbani *ICISSP 2018*	80+ bidirectional statistical flow features & multi-family realistic attack profiles.	`datasets/` `app/services/dataset_streamer.py`
3. Isolation Forest **Streaming Anomaly Detection**	Liu, Ting, Zhou *IEEE ICDM 2008*	Partitioning tree path lengths to isolate high-entropy outliers without labeled priors.	`app/detection/` `app/services/knowledge_service.py`
4. RAG Framework **Retrieval-Augmented Generation**	Lewis et al. *NeurIPS 2020*	Grounded LLM security responses via non-parametric incident & playbook retrieval.	`app/services/ai_service.py` `app/services/knowledge_service.py`

5. ML Benchmarks
RF, XGBoost, CNN, LSTM
	Breiman (2001), Chen (2016)
	KDD & ML Journals
		Comparative evaluation under identical zero-day held-out class splits.	`ml/random_forest/`,
			`ml/xgboost/`
			`ml/cnn/`, `ml/lstm/`

6. NIST SP 800-30 & MITRE **Risk & Kill-Chain Graph**	NIST Special Pub (2012) Strom et al. (MITRE 2018)	Multi-hop alert correlation & dynamic quantitative incident risk scoring (0-100).	`app/risk/risk_engine.py` `components/graph/`
7. Sigma Standard **Generic SIEM Rule Engine**	Roth & Patzke *Sigma Standard (2017)*	AST compilation of YAML rule trees matching high-velocity Redis stream events in <1ms.	`app/detection/rule_engine.py` `app/workers/event_worker.py`

2. Deep Dive: GraphSAGE for Zero-Day Attack Discovery

Reference Paper: Hamilton, W. L., Ying, R., & Leskovec, J. (2017). *Inductive representation learning on large graphs.* Advances in Neural Information Processing Systems (NeurIPS 2017), pp. 1024-1034.

2.1 The Theoretical Cyber Problem: Why Standard ML Fails on Zero-Days

Standard machine learning models (Random Forest, XGBoost, Deep Neural Networks) assume that network traffic records are independent, identically distributed (i.i.d.) tabular vectors. In real-world enterprise environments, attackers exploit novel zero-day vulnerabilities whose payload signatures and packet size distributions have never been seen. Consequently, supervised classifiers fail because they only recognize the exact feature distributions present in their training set.

However, cyber attacks cannot escape **graph topological constraints**. Whether an exploit is known or zero-day, the adversary must perform network reconnaissance, establish command-and-control (C2) beaconing, sweep subnets, or execute lateral movement. These behaviors form distinct, identifiable relational patterns across the communication topology.

2.2 Mathematical Formulation & Inductive Neighborhood Aggregation

Unlike transductive Graph Convolutional Networks (GCN) which require the full graph Laplacian matrix at train time and cannot generalize to unseen nodes, **GraphSAGE** learns parameterized aggregation functions that inductively generate embeddings for new, unobserved nodes. At step k, the node representation $h_v^{(k)}$ is updated via:

GraphSAGE Neighborhood Aggregation Formulation:

$$\mathbf{h}_{N(v)}^{(k)} = \text{AGGREGATE}_k \left(\{ \mathbf{h}_u^{(k-1)}, \forall u \in N(v) \} \right)$$

$$\mathbf{h}_v^{(k)} = \sigma \left(\mathbf{W}^{(k)} \cdot [\mathbf{h}_v^{(k-1)} \| \mathbf{h}_{N(v)}^{(k)}] \right)$$

where $\|$ represents vector concatenation, $\mathbf{W}^{(k)}$ is the trainable layer weight, σ is the non-linear ReLU activation, and AGGREGATE is Mean Aggregation across the sampled neighborhood $N(v)$.

2.3 Concrete Implementation in SentinelX

- **Topological Graph Construction (ml/graphsage/graph_construction.py):** Transforms high-dimensional tabular flow vectors into a topological graph. In the default configuration, nodes represent individual IP endpoints or flow vectors, and edges are dynamically formed via *k-Nearest Neighbors (k-NN)* cosine similarity over scaled flow features. This links latent attack communications across endpoints.
- **PyTorch Geometric Neural Architecture (ml/graphsage/model.py):** Constructed using torch_geometric.nn.SAGEConv in a 2-layer inductive architecture:
  Layer 1: SAGEConv(in_channels=Flow Features, hidden_channels=32) → ReLU → Dropout(p=0.2)
  Layer 2: SAGEConv(hidden_channels=32, out_channels=2) → Softmax (Benign vs Malicious).
- **The Zero-Day Held-Out Class Protocol (ml/preprocessing/pipeline.py::zero_day_split):** To evaluate true zero-day performance, SentinelX completely strips an entire attack class (e.g., *PortScan*, *DoS Slowloris*, or *SQL Injection*) from the training set. The model is trained strictly on benign traffic and other attack types. At test time, the novel unseen attack is presented. GraphSAGE consistently outperforms classical models because neighbor aggregation reveals the structural anomaly of the exploit.

3. CIC-IDS2017 & CSE-CIC-IDS2018: Threat Telemetry & Streaming

Reference Paper: Sharafaldin, I., Lashkari, A. H., & Ghorbani, A. A. (2018). *Toward generating a new intrusion detection dataset and analysis of malicious traffic.* Proceedings of the 4th International Conference on Information Systems Security and Privacy (ICISSP 2018), pp. 108-116.

Academic Significance: Created by the Canadian Institute for Cybersecurity (University of New Brunswick), CIC-IDS2017 resolves the critical weaknesses of legacy datasets (KDD99, DARPA98) which lacked modern protocol diversity, encrypted traffic, and contemporary attack vectors. CIC-IDS captures 7 complete attack families across 80+ statistical flow metrics.

Implementation in SentinelX:

- **Feature Normalization (backend/app/services/dataset_service.py):** Normalizes 80+ bidirectional flow metrics: Flow Duration, Flow IAT Mean/Std, Fwd/Bwd Packet Lengths, Flag counts (PSH, ACK, FIN, URG, SYN), Window Sizes, and Subflow metrics.
- **Real-Time Redis Streamer (backend/app/services/dataset_streamer.py):** Replays packets into Redis Streams (sentinelx:events) at configurable rates (1-100 EPS) with smart offset skipping to eliminate benign delays, allowing live detection demonstration of PortScan, DoS, SQLi, and Brute Force attacks.

4. Isolation Forest for Streaming Anomaly Detection

Reference Paper: Liu, F. T., Ting, K. M., & Zhou, Z. H. (2008). *Isolation Forest.* Eighth IEEE International Conference on Data Mining (ICDM 2008), pp. 413-422. IEEE.

Algorithmic Rationale: Unlike density estimation (LOF) or distance-based clustering (DBSCAN) which require $O(n^2)$ comparisons, Isolation Forest isolates outliers by building an ensemble of completely random binary trees (*iTrees*). Because anomalies have distinct attribute values, they are isolated at significantly shallower tree depths (shorter path lengths $h(x)$).

Isolation Forest Anomaly Scoring Formula:

$$s(x, n) = 2^{- E(h(x)) / c(n)}$$

where $E(h(x))$ is the average path length across the tree ensemble, and $c(n) = 2\ ln(n - 1) + 0.5772156649 - 2(n - 1)/n$ is the average path length of an unsuccessful binary search tree.
Scores close to 1 indicate confirmed anomalies; scores < 0.5 indicate normal traffic.

SentinelX Implementation (backend/app/detection/): Isolation Forest acts as SentinelX's unsupervised frontline defense. Any high-entropy packet burst or evasion attempt that avoids static signature rules triggers an anomaly score $s \geq 0.65$, escalating the event directly into the SOC incident graph for automated investigation.

5. Retrieval-Augmented Generation (RAG) for SOC Operations

Reference Paper: Lewis, P., Perez, E., Piktus, A., et al. (2020). *Retrieval-augmented generation for knowledge-intensive NLP tasks.* Advances in Neural Information Processing Systems (NeurIPS 2020), 33, 9459-9474.

The Problem in Autonomous SOCs: Generative AI assistants in cybersecurity frequently suffer from catastrophic hallucination, inventing non-existent IP addresses, referencing inaccurate CVEs, or suggesting dangerous firewall configurations.

SentinelX Implementation (backend/app/services/ai_service.py): SentinelX strictly implements RAG by coupling parametric LLM capability with non-parametric document and telemetry retrieval:
1. **Real-time Telemetry Ingestion:** Live incident reports, attacker IPs, affected hostnames, and rule matches are indexed into memory.
2. **MITRE ATT&CK; & Playbook Grounding:** Enterprise mitigation strategies and firewall containment playbooks are dynamically retrieved.
3. **Enforced Grounding Constraint:** The AI Copilot is instructed to formulate remediation responses exclusively using retrieved facts.

6. Standards: NIST SP 800-30 & Sigma Rule AST Engine

6.1 NIST SP 800-30 Quantitative Incident Risk Scoring (backend/app/risk/risk_engine.py)

Following the risk assessment guidelines in *NIST Special Publication 800-30 Revision 1*, SentinelX replaces static severity tags with dynamic quantitative risk scores ranging from 0 to 100:

SentinelX Quantitative Risk Calculation:

RiskScore = min(100, max(0, BaseSeverityWeight + min(AlertCount × 5, 25) + MITRE_Bonus))

where BaseSeverity ∈ {Critical: 50, High: 35, Medium: 20, Low: 10}, multi-event aggregation scales with incident frequency, and confirmed MITRE technique mapping awards +15 points.

6.2 Sigma Generic SIEM Signature AST Engine (backend/app/detection/rule_engine.py)

Based on the open standard by Florian Roth and Thomas Patzke, SentinelX compiles YAML-based Sigma rules into an optimized Abstract Syntax Tree (AST) in Python. The background event worker evaluates incoming Redis Stream events against the rule AST in sub-millisecond execution time, instantly flagging port scans, remote exposure, and credential access.

7. End-to-End Architectural Integration

All academic research components operate in a unified, multi-tiered pipeline inside SentinelX:

Architecture Layer	Academic Component	Production Function in SentinelX
1. Telemetry Ingestion	CIC-IDS2017 & psutil Sockets	Dual-mode ingestion: live host connections (mode='LIVE') and dataset replay (mode='DATASET').
2. Stream Broker	Redis Streams (XADD / XREADGROUP)	Persistent, decoupled streaming broker with auto-flush on mode switches.
3. Detection Worker	Sigma AST & Isolation Forest	Evaluates stream events in sub-millisecond time; flags known rules and high-entropy anomalies.
4. Incident Correlation	NIST SP 800-30 & MITRE Graph	Correlates isolated alerts by IP into cohesive attack kill-chains with 0-100 dynamic risk scores.
5. Zero-Day GNN	GraphSAGE (PyTorch Geometric)	Inductive neighbor aggregation detecting unclassified, completely held-out attack vectors.
6. Copilot & Intelligence	RAG (Retrieval-Augmented Gen)	Grounds AI response strictly in real retrieved incident telemetry and containment playbooks.
7. Endpoint SOAR	Automated Incident Response	Dispatches remote containment commands (Windows Firewall IP block, taskkill) back to the agent.

8. Formal Academic Bibliography & References

[1] Hamilton, W. L., Ying, R., & Leskovec, J. (2017). Inductive representation learning on large graphs. *Advances in Neural Information Processing Systems (NeurIPS 2017)*, 30, 1024-1034.

[2] Sharafaldin, I., Lashkari, A. H., & Ghorbani, A. A. (2018). Toward generating a new intrusion detection dataset and analysis of malicious traffic. *Proceedings of the 4th International Conference on Information Systems Security and Privacy (ICISSP 2018)*, pp. 108-116.

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[6] Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5-32.

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[8] Strom, B. E., Applebaum, A., Miller, D. P., et al. (2018). MITRE ATT&CK;: Design and Philosophy. *Technical Report MTR180126*, The MITRE Corporation.

[9] Roth, F., & Patzke, T. (2017). Sigma: Generic Signature Format for SIEM Systems. *Sigma Project Specification*.